Title: Washroom Availability Indication System

By:

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1. Abstract

As the number of hostel inmates increase, the issue of availability of washrooms during peak hours is a major issue that is faced by the hostellers. As a result we are inspired to develop this application that helps to resolve this issue and ease the life of the residents on a daily basis. An easy to use user friendly application that keeps track of the availability of restrooms will help save time and allow the user to find an empty cubicle with ease and also eliminate any sort of waiting period to avail the restroom services**.** The washroom availability system allows efficient usage of toilets, display of empty and filled cubicles. This proposed system is especially designed for hostels/ buildings with common washroom areas. The app would also provide direct root access to the building administrator so they could update if any cubicle(s) are out of service.

2. Introduction

The advantages of the washroom availability system are efficient usage of toilets, display of empty and filled cubicles. This proposed system is especially designed for hostels/ buildings with common washroom areas. The app would also provide direct root access to the building administrator so they could update if any cubicle(s) are out of service.

Whenever a resident wants to go to the washroom he/she can check the availability of the cubicle beforehand hence saving the time for the person by redirecting directly to another floor if required when the cubicle is not available on their desired floor.

3. Existing Systems

Our model is based on the idea similar to automated parking management system.It includes and updates the information on the app whether a free space is available in the area.On the basis of this idea we have also added the option to disable a free space if that cubicle is out of service which can be set by the admin of the system only.

4. Problem Statement

We were motivated to create this application since it would assist in resolving the issue of availability of washrooms and make everyday life easier for the residents as the number of hostel inmates rises and the availability of washrooms during peak hours becomes a major problem for the hostellers.

5. Introduction to Proposed Model

Our projects include concepts from multiple domains such as Internet of Things ,Database Management Systems , App Development ,UX/UI Design and working with sensors. The main focus of the project will be based on the working application that the user interacts with .Designing and developing the application in a manner that follows the design principles of human computer interaction that will allow a smooth and efficient interaction between application and user will be a topmost priority.

6. Proposed Solution

An easy-to-use user-friendly application that keeps track of the availability of restrooms will help save time and allow the user to find an empty cubicle with ease and also eliminate any sort of waiting period to avail the restroom services.

The proposed system is especially designed for hostels/ buildings with common washroom areas. The app would also provide direct root access to the building administrator so they could update if any cubicle(s) are out of service.

Main goal would be to enhance user experience while searching for empty cubicles. And also an overall view for the admin to see the status of each cubicle in the whole building.

7. Components of Model

7.1 Hardware:

1. IR Sensors
2. Arduino Uno
3. NodeMcu esp8266
4. Jumper wires
5. Cardboard(for 3d Model)
6. Phone

7.2 Software:

1. Android Studio(Java/Volley)
2. ThingSpeak

8. Description of the components (HW / SW / Tech. Specifications)

The Arduino UNO microcontroller is the brains of the model, collecting the data from all the Infrared sensors used, which tell if the door is closed by using IR waves to tell if there is an obstruction which triggers the sensor.

To establish connection to a web server, we have used the ESP8266 module which is connected to the Arduino via serial communication and uploads all the information to the web database (Thingspeak web database).

The app we've made reads from the Thingspeak database via APIs and reflects it on the UI.

9. Block Diagram

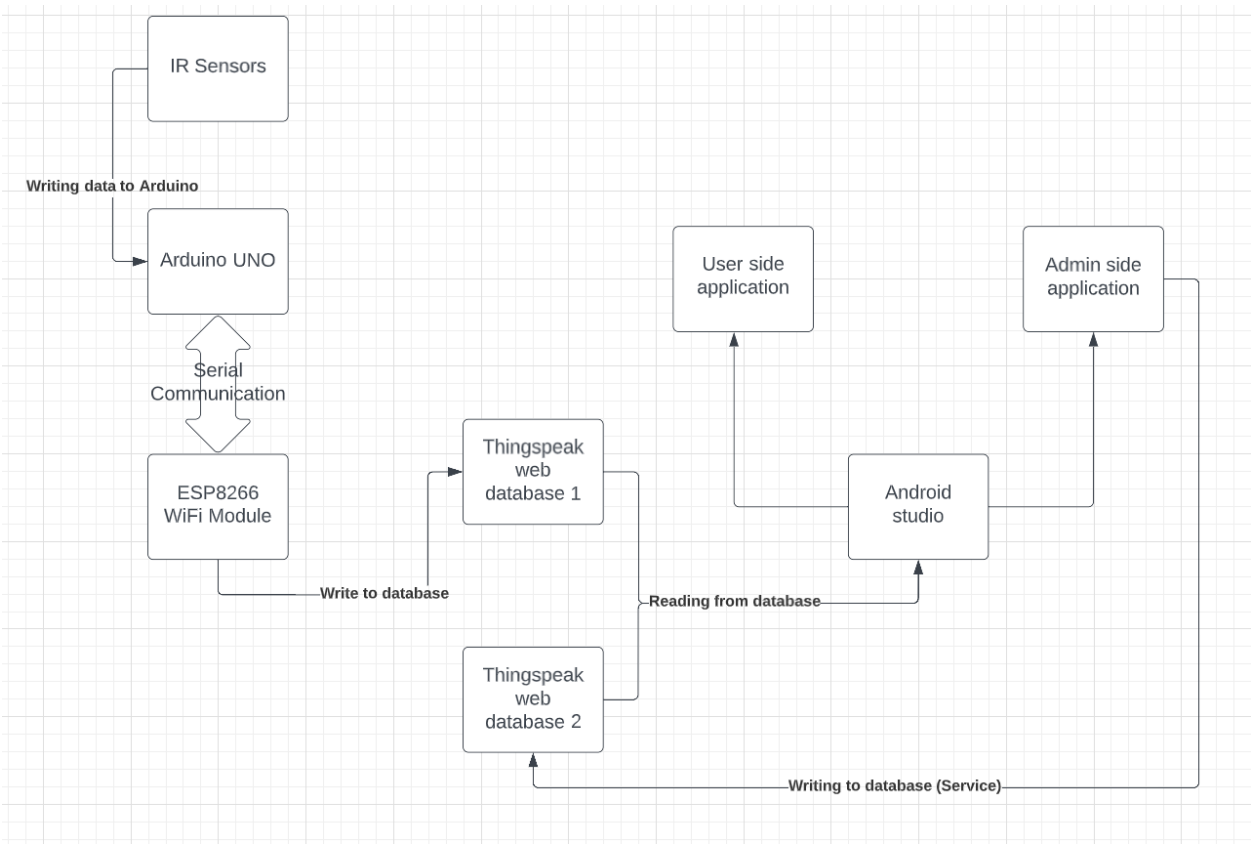


Fig 1

10. Working Mechanism

When the hatchbolt of the door closes the IR Sensor will sense it.When it is closed, ‘0’ will be sent to the server.Serial communication takes place between the arduino board and nodeMcu.Then the data is posted to server(channel-washroom) as a JSON object.The data of all the IR sensor are posted at once.

The app will continuously read the data from the server.When the data is ‘1’ the cubicle will be shown vacant.When the data is ‘0’ the cubicle will be shown occupied.

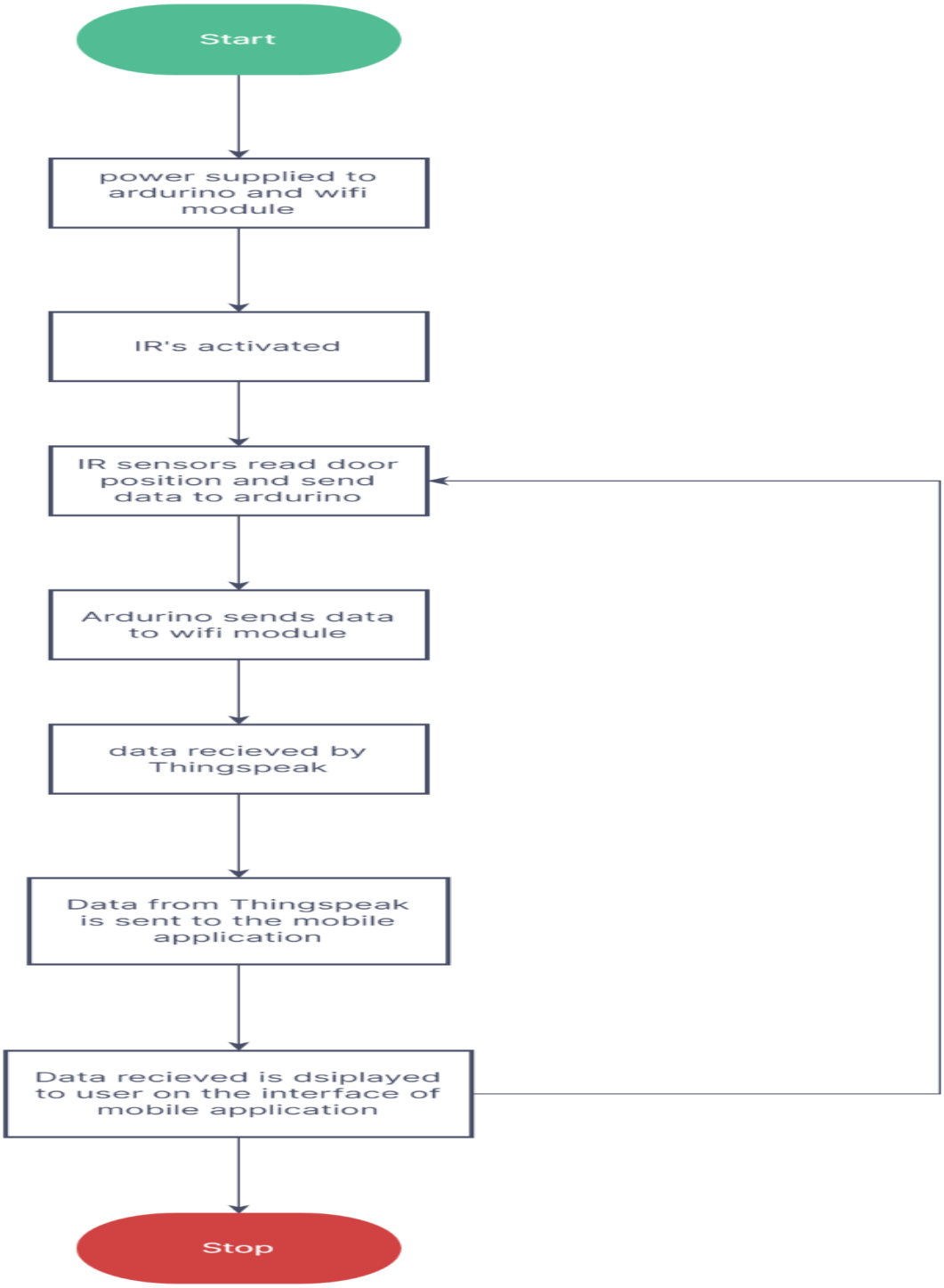
Now if the admin sets the status of a cubicle as out of service.A JSON object will be sent to the server (channel-washroom_admin).If the data is ‘0’ then the washroom will be showed Out of service and if ‘1’ then the washroom will be showed as ‘In service’.

The admin has same functions as user.It also has the option to set the status of washroom.When the All info button is created, the info about each and every washroom will be shown in a tabular form.If one cubicle is out of service,that floor will be highlighted as grey.If both the cubicles are out of service then the floor will be highlighted as red.

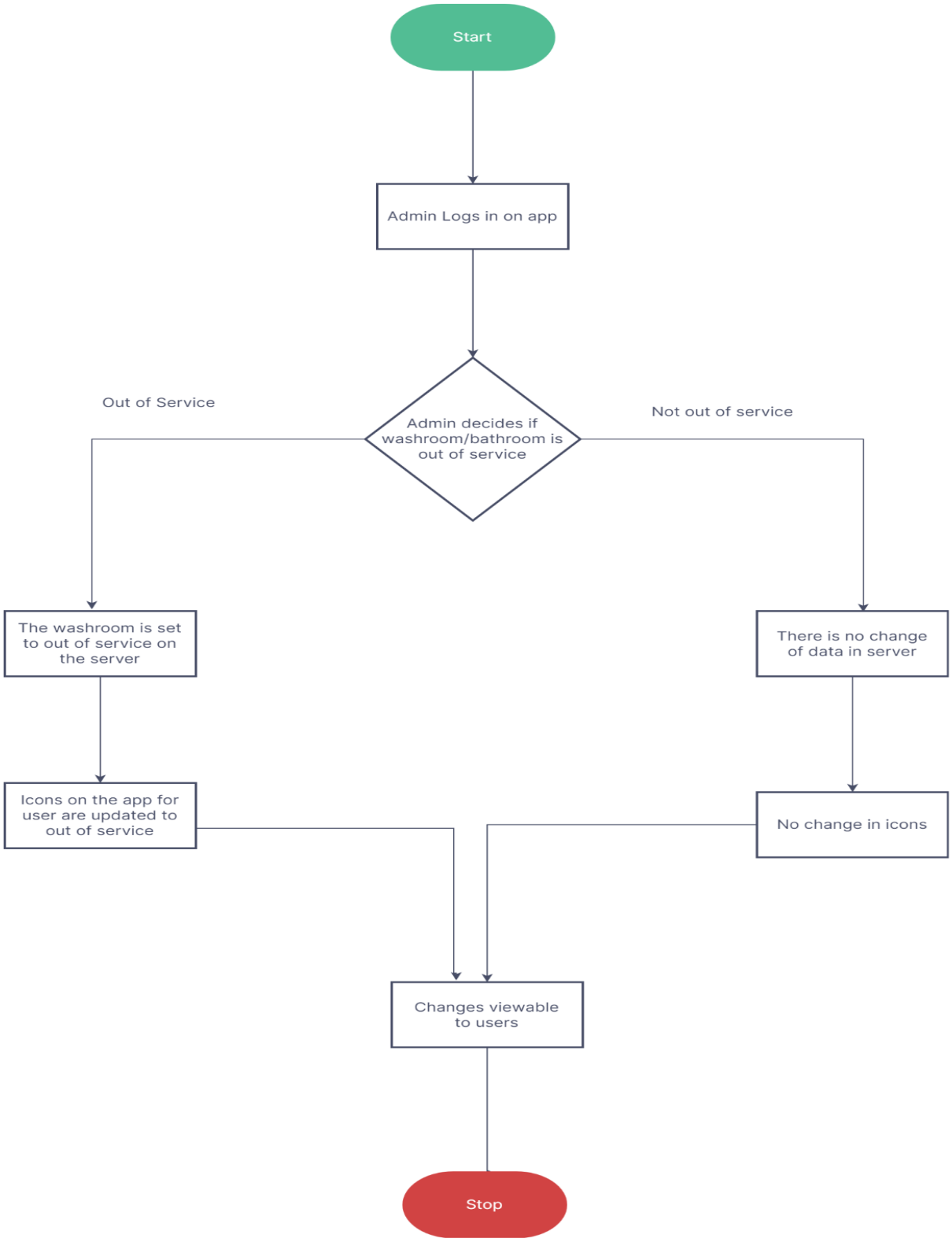
The user when opens the app all the data will be read from server.When the user selects the floor the status of each cubicle will be shown.If no washroom on the floor is available then a button will be shown on clicking which the nearest washroom available will be shown and will be redirected to that floor.If no washroom is available then a toast message will be shown saying “No Washroom Available”. Similarly there is a button to show Nearest Bathroom.

11. Flow Diagram / Flow Chart

11.1 User application flow



11.2 Admin application flow



12. Code (Input / Execution / Output)

12.1 ThingSpeak

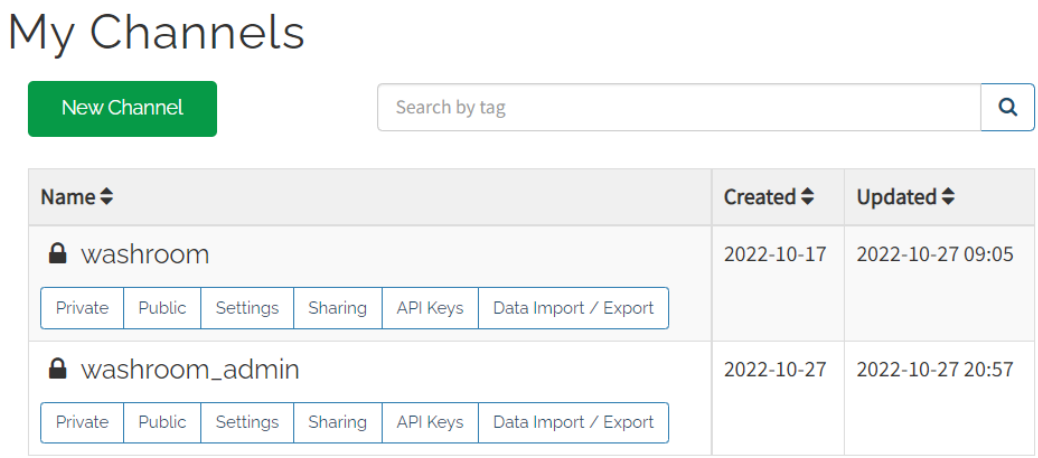


Fig 2-ThingSpeak image

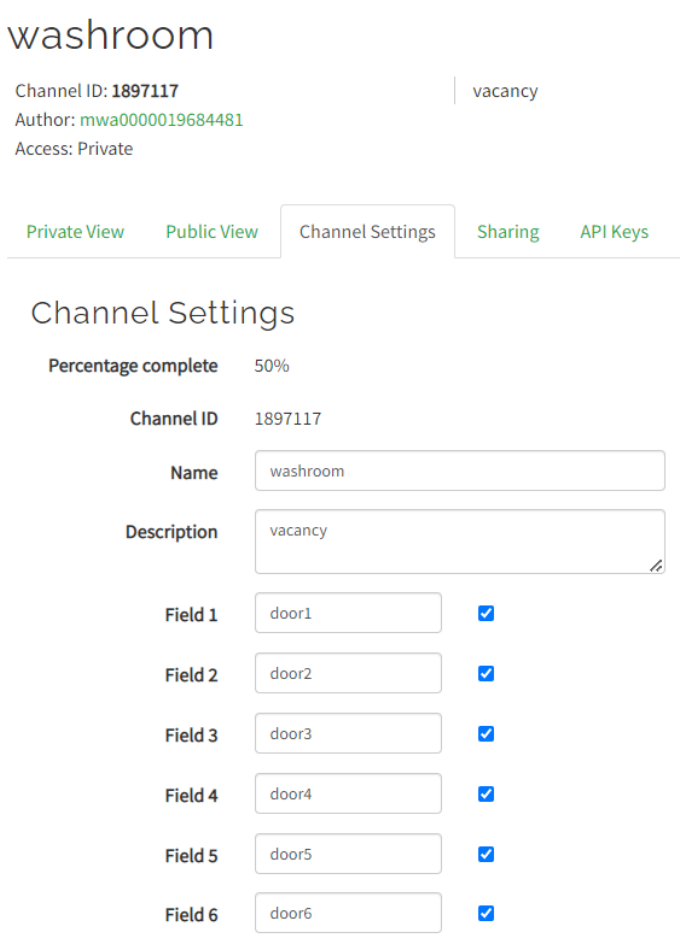


Fig 3-ThingSpeak image

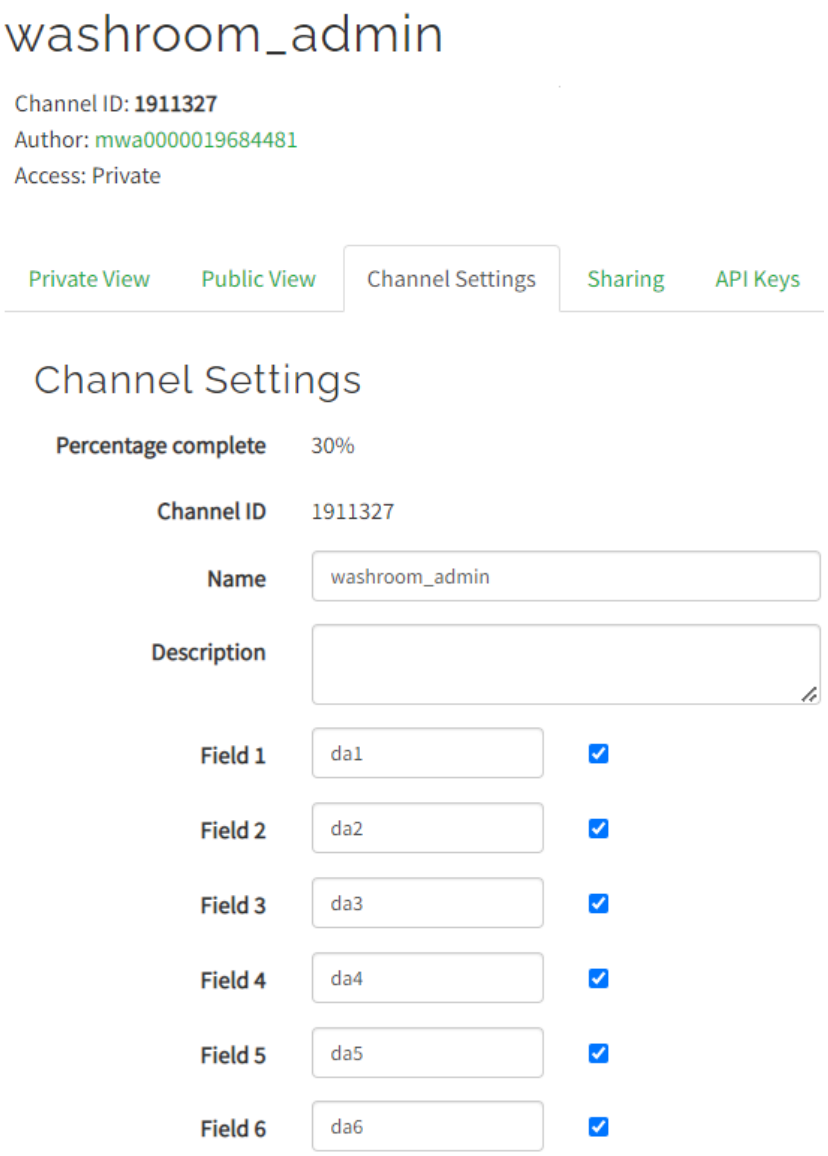


Fig-4 ThingSpeak image

12.2.1 Arduino

arduino §

```
#include <SoftwareSerial.h>
#include <ArduinoJson.h>
SoftwareSerial nodemcu(5,6);
// 5=Rx 6 =Tx
String str;
void setup() {
  // put your setup code here, to run once:
  Serial.begin(9600);
  delay(1000);
  Serial.println("Program Started");
  nodemcu.begin(9600);
  pinMode(7,INPUT);
  pinMode(8,INPUT);
  pinMode(9,INPUT);
  pinMode(10,INPUT);
  pinMode(11,INPUT);
  pinMode(12,INPUT);
}
void loop() {
  // put your main code here, to run repeatedly:
  int a=digitalRead(7),b=digitalRead(8);
  int c=digitalRead(9),d=digitalRead(10);
  int e=digitalRead(11),f=digitalRead(12);
  Serial.print("IR 1:");
  Serial.print(a);
```

```
  Serial.print(" IR 2:");
  Serial.println(b);
    Serial.print("IR 3:");
  Serial.print(c);
  Serial.print(" IR 4:");
  Serial.println(d);
    Serial.print("IR 5:");
  Serial.print(e);
  Serial.print(" IR 6:");
  Serial.println(f);
  StaticJsonBuffer<1000> jsonBuffer;
  JsonObject& data = jsonBuffer.createObject();
  data["IR1"]=a;
  data["IR2"]=b;
   data["IR3"]=c;
  data["IR4"]=d;
   data["IR5"]=e;
  data["IR6"]=f;
  data.printTo(nodemcu);
  jsonBuffer.clear();
  Serial.println("Data Sent!");
  delay(1000);
}
```

12.2.2 NodeMCU

connect_wifi_to_nodemcu

```
#include <ESP8266WiFi.h>
#include "ThingSpeak.h"

const char *ssid =  "HARSH 7235";
const char *pass =  "P660e4<8";

#include <SoftwareSerial.h>
#include <ArduinoJson.h>

WiFiClient client;

unsigned long myChannelNumber=1;
const char* myWriteAPIKey="G27P24W5FVDW9OJZ";
SoftwareSerial nodemcu(D6, D5);
//D6-Rx d5 -Tx
void setup()
{
        Serial.begin(9600);
        delay(10);
        nodemcu.begin(9600);
        while (!Serial) continue;
        Serial.println("Connecting to ");
        Serial.println(ssid);

        WiFi.begin(ssid, pass);
```

```
        while (WiFi.status() != WL_CONNECTED)
            {
              delay(500);
              Serial.print(".");
            }
       Serial.println("");
       Serial.println("WiFi connected");
       WiFi.mode(WIFI_STA);
       ThingSpeak.begin(client);
}

void loop()
{
  StaticJsonBuffer<1000> jsonBuffer;
  JsonObject& data = jsonBuffer.parseObject(nodemcu);
  if (data == JsonObject::invalid()) {
    //Serial.println("Invalid Json Object");
    jsonBuffer.clear();
    return;
  }
  Serial.println("JSON Object Recieved");
  int a = data["IR1"];
  int b = data["IR2"];
  int c = data["IR3"];
  int d = data["IR4"];
  int e = data["IR5"];
  int f = data["IR6"];
  Serial.println("----------------------------------------");
  Serial.print("IR1 :");Serial.print(a);
  Serial.print("IR2 :");Serial.println(b);
  Serial.print("IR3 :");Serial.print(c);
  Serial.print("IR4 :");Serial.println(d);
  Serial.print("IR5 :");Serial.print(e);
  Serial.print("IR6 :");Serial.println(f);
   ThingSpeak.setField(1, a);
   ThingSpeak.setField(2, b);
    ThingSpeak.setField(3, c);
   ThingSpeak.setField(4, d);
    ThingSpeak.setField(5, e);
   ThingSpeak.setField(6, f);
   ThingSpeak.writeFields(myChannelNumber, myWriteAPIKey);
   Serial.println("ths");

   //ThingSpeak.writeField(myChannelNumber, 1, a, myWriteAPIKey);
   //ThingSpeak.writeField(myChannelNumber, 2, b, myWriteAPIKey);
  delay(100);
}
```

12.3 App Development

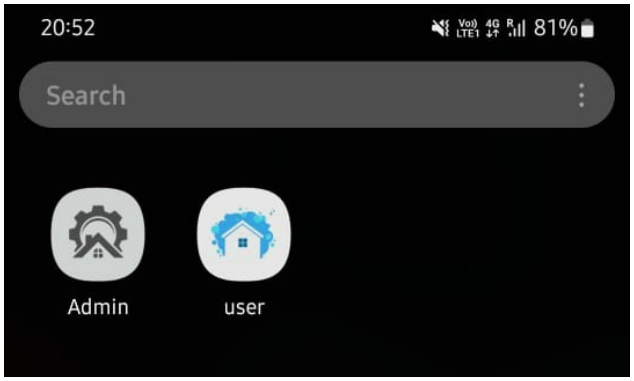


Fig 5-App Image

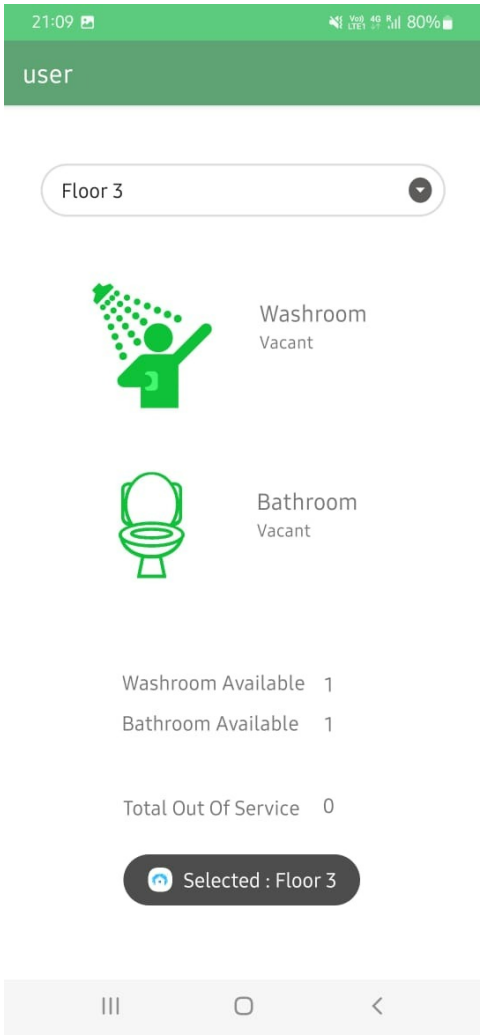


Fig 6- App Image

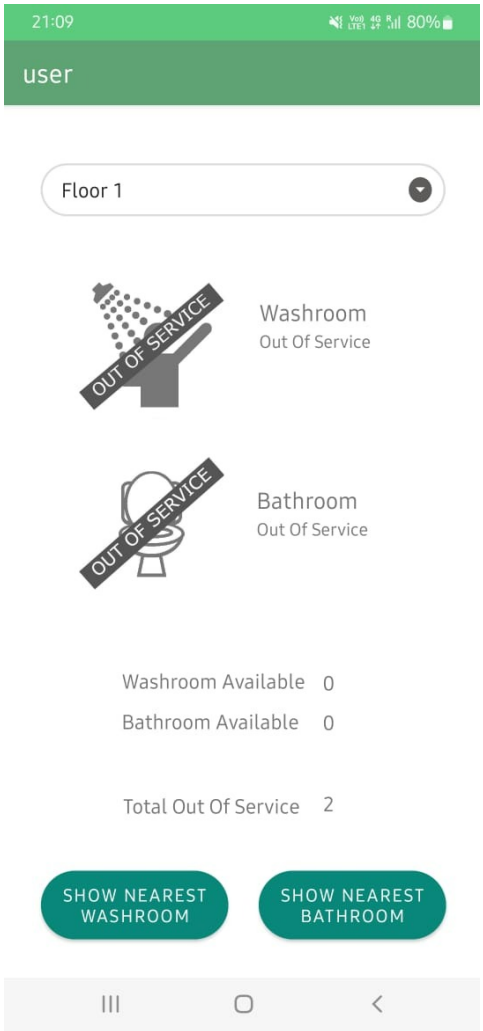


Fig 7-App Image

12.4 Android Studio (User-App)

```
package com.rixxx.user1;

import androidx.annotation.NonNull;
import androidx.annotation.RequiresApi;
import androidx.appcompat.app.AppCompatActivity;
import android.graphics.Color;
import android.os.Build;
import android.os.Bundle;
import android.os.Handler;
import android.util.Log;
import android.view.View;
import android.view.ViewGroup;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.CompoundButton;
import android.widget.ImageView;
import android.widget.Spinner;
import android.widget.Switch;
import android.widget.TextView;
import android.widget.Toast;
import java.text.BreakIterator;
import java.util.ArrayList;
import java.util.Arrays;
import java.util.HashMap;
import java.util.List;
import java.util.Map;
import com.android.volley.Request;
import com.android.volley.RequestQueue;
import com.android.volley.Response;
import com.android.volley.VolleyError;
import com.android.volley.VolleyLog;
import com.android.volley.toolbox.JsonObjectRequest;
import com.android.volley.toolbox.StringRequest;
import com.android.volley.toolbox.Volley;
import org.json.JSONException;
import org.json.JSONObject;
```

```
public class MainActivity extends AppCompatActivity {

    TextView textView1;
    TextView textView2;
    Button b1,b2;
    int count=0;
    TextView wa,ba,ou;
    RequestQueue queue,queue1,queue2;
    ImageView im1,im2;
    String d1,d2,d3,d4,d5,d6;
    Spinner spinner;
    String dd1,dd2,dd3,dd4,dd5,dd6;
    String da1,da2,da3,da4,da5,da6;
    String arr[]=new String[6];
    String URL =
"https://api.thingspeak.com/channels/1897117/feeds.json?api_key=1L024NOIEU
XXUVSB&results=2";
    @RequiresApi(api = Build.VERSION_CODES.N)
    //adminside
    String
url2="https://api.thingspeak.com/channels/1911327/feeds.json?api_key=B1B7N
XKTIA3LQGK8&results=2";
    String
url="https://api.thingspeak.com/update?api_key=0MSY5CSW5B65MB5G";
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        content();
        b1= findViewById(R.id.buttonw);
        b2= (Button)findViewById(R.id.buttonb);
        b1.setEnabled(false);
        b1.setVisibility(View.INVISIBLE);
        b2.setEnabled(false);
        b2.setVisibility(View.INVISIBLE);
       spinner = findViewById(R.id.spinner);
           String[] plants = new String[]{
                "Select Floor...",
                "Floor 1",
                "Floor 2",
```

```
        "Floor 3",

};
List<String> plantsList = new ArrayList<>
        (Arrays.asList(plants));
ArrayAdapter<String> spinnerArrayAdapter
        = new ArrayAdapter<String>(
        this,
        android.R.layout.simple_dropdown_item_1line,
        plantsList
){

    @Override
    public boolean isEnabled(int position){
        // Disable the first item from Spinner
        // First item will be use for hint
        //need to be updated continuosly
        setArray();
        return position != 0;

    }
    @Override
    public View getDropDownView(
            int position, View convertView,
            @NonNull ViewGroup parent) {
        // Get the item view
        View view = super.getDropDownView(
                position, convertView, parent);
        TextView textView = (TextView) view;

        if(position == 0){
            // Set the hint text color gray
            textView.setTextColor(Color.GRAY);

        }
        return view;
    }
};

// Set the drop down view resource
```

```
spinnerArrayAdapter.setDropDownViewResource(
        android.R.layout.simple_dropdown_item_1line
);

// Spinner on item selected listener
spinner.setOnItemSelectedListener(
        new AdapterView.OnItemSelectedListener() {
            @Override
            public void onItemSelected(
                    AdapterView<?> parent, View view,
                    int position, long id) {

                // Get the spinner selected item text
                String selectedItemText = (String) parent
                        .getItemAtPosition(position);

                // If user change the default selection
                // First item is disable and
                // it is used for hint
                textView1 = findViewById(R.id.info1);
                textView2 = findViewById(R.id.info2);
                wa = findViewById(R.id.wav);
                ba = findViewById(R.id.bav);
                ou = findViewById(R.id.ous);

                //floor1

                if(position == 1){
                   // Notify the selected item text
                    Toast.makeText(
                            getApplicationContext(),
                            "Selected : "
                                    + selectedItemText,
                            Toast.LENGTH_SHORT).show();
                    d1=setCheck(da1,dd1);
                    d2=setCheck(da2,dd2);
                    d3=setCheck(da3,dd3);
                    d4=setCheck(da4,dd4);
                    d5=setCheck(da5,dd5);
                    d6=setCheck(da6,dd6);
```

```
        String print1=setInfo(d1);
        String print2=setInfo(d2);
        //need to be updated
        textView1.setText(print1);
        textView2.setText(print2);
        int count=0;
        if(d1.equals("3"))
        {
            count+=1;

            wa.setText("0");
        }else{
            wa.setText(d1);

        }

        if(d2.equals("3"))
        {
            count+=1;

            ba.setText("0");
        }else{
            ba.setText(d2);

        }
        ou.setText(""+count);
        setButton(d1,d2,position);
        setImage(d1,d2);

    }//floor2
    else  if(position == 2){
         // Notify the selected item text
        Toast.makeText(
                getApplicationContext(),
                "Selected : "
                        + selectedItemText,
                Toast.LENGTH_SHORT).show();
        d1=setCheck(da1,dd1);
        d2=setCheck(da2,dd2);
        d3=setCheck(da3,dd3);
```

```
        d4=setCheck(da4,dd4);
        d5=setCheck(da5,dd5);
        d6=setCheck(da6,dd6);
        String print1=setInfo(d3);
        String print2=setInfo(d4);
        //need to be updated
        textView1.setText(print1);
        textView2.setText(print2);
        int count=0;
        if(d3.equals("3"))
        {
            count+=1;

            wa.setText("0");
        }else{
            wa.setText(d3);

        }

        if(d4.equals("3"))
        {
            count+=1;

            ba.setText("0");
        }else{
            ba.setText(d4);

        }
        ou.setText(""+count);
        setImage(d3,d4);
        setButton(d3,d4,position);

    }//floor3
    else  if(position == 3){
      // Notify the selected item text
        Toast.makeText(
                getApplicationContext(),
                "Selected : "
                        + selectedItemText,
                Toast.LENGTH_SHORT).show();
```

```
            d1=setCheck(da1,dd1);
            d2=setCheck(da2,dd2);
            d3=setCheck(da3,dd3);
            d4=setCheck(da4,dd4);
            d5=setCheck(da5,dd5);
            d6=setCheck(da6,dd6);
            String print1=setInfo(d5);
            String print2=setInfo(d6);
            //need to be updated
            textView1.setText(print1);
            textView2.setText(print2);
            int count=0;
            if(d5.equals("3"))
            {
                count+=1;

                wa.setText("0");
            }else{
                wa.setText(d5);

            }

            if(d6.equals("3"))
            {
                count+=1;

                ba.setText("0");
            }else{
                ba.setText(d6);

            }
            ou.setText(""+count);
            setImage(d5,d6);
            setButton(d5,d6,position);

        }
    }
    @Override
```

```
                    public void onNothingSelected(
                            AdapterView<?> parent) {

                    }
                });
        // Finally, data bind the spinner object with adapter
        spinner.setAdapter(spinnerArrayAdapter);
        contData();
    }
    private void setArray() {
        arr[0]=da1;
        arr[1]=da2;
        arr[2]=da3;
        arr[3]=da4;
        arr[4]=da5;
        arr[5]=da6;
    }
    public void content() {
        queue = Volley.newRequestQueue(this);
        StringRequest request = new StringRequest(Request.Method.GET, URL,
new Response.Listener<String>() {
            @Override
            public void onResponse(String response) {

                String s=response.toString();
                dd1=Character.toString(s.charAt(s.length()-70));
                dd2=Character.toString(s.charAt(s.length()-57));
                dd3=Character.toString(s.charAt(s.length()-44));
                dd4=Character.toString(s.charAt(s.length()-31));
                dd5=Character.toString(s.charAt(s.length()-18));
                dd6=Character.toString(s.charAt(s.length()-5));

            }
        }, new Response.ErrorListener() {
            @Override
            public void onErrorResponse(VolleyError error) {
                Log.d("error",error.toString());
            }
        });
        queue.add(request);
```

```
        //adminside
        queue1 = Volley.newRequestQueue(this);
        StringRequest request1 = new StringRequest(Request.Method.GET,
url2, new Response.Listener<String>() {
            @Override
            public void onResponse(String response) {

                String sk=response.toString();
                da1=Character.toString(sk.charAt(sk.length()-70));
                da2=Character.toString(sk.charAt(sk.length()-57));
                da3=Character.toString(sk.charAt(sk.length()-44));
                da4=Character.toString(sk.charAt(sk.length()-31));
                da5=Character.toString(sk.charAt(sk.length()-18));
                da6=Character.toString(sk.charAt(sk.length()-5));

            }
        }, new Response.ErrorListener() {
            @Override
            public void onErrorResponse(VolleyError error) {
                Log.d("error",error.toString());
            }
        });
        queue1.add(request1);
        refresh(1000);
    }

    private void refresh(int i) {
        final Handler handler=new Handler();
        final Runnable runnable=new Runnable() {
            @Override
            public void run() {

                content();
                setArray();
                contData();
                // finalData();
            }
        };
```

```
        handler.postDelayed(runnable,i);
    }

    private void setAdminData(String f1, String dad1, String dad2) {
        if (f1.equals("1"))
        {
            arr[0]=dad1;
            arr[1]=dad2;
        }else if(f1.equals("2")){
            arr[2]=dad1;
            arr[3]=dad2;
        }else if(f1.equals("3")){
            arr[4]=dad1;
            arr[5]=dad2;
        }
    }



    private void contData(){
        int pos=spinner.getSelectedItemPosition();
        textView1 = findViewById(R.id.info1);
        textView2 = findViewById(R.id.info2);
        wa = findViewById(R.id.wav);
        ba = findViewById(R.id.bav);
        ou = findViewById(R.id.ous);
        //floor1
        if(pos == 1){
            d1=setCheck(da1,dd1);
            d2=setCheck(da2,dd2);
            d3=setCheck(da3,dd3);
            d4=setCheck(da4,dd4);
            d5=setCheck(da5,dd5);
            d6=setCheck(da6,dd6);
            //need to be updated
            String print1=setInfo(d1);
            String print2=setInfo(d2);
            //need to be updated
            textView1.setText(print1);
            textView2.setText(print2);
```

```
        int count=0;
        if(d1.equals("3"))
        {   count+=1;
            wa.setText("0");
        }else{
            wa.setText(d1);
        }
        if(d2.equals("3"))
        {   count+=1;
            ba.setText("0");
        }else{
            ba.setText(d2);
        }
        ou.setText(""+count);
        setButton(d1,d2,pos);
        setImage(d1,d2);
    }//floor2
    else  if(pos == 2){
        d1=setCheck(da1,dd1);
        d2=setCheck(da2,dd2);
        d3=setCheck(da3,dd3);
        d4=setCheck(da4,dd4);
        d5=setCheck(da5,dd5);
        d6=setCheck(da6,dd6);
        String print1=setInfo(d3);
        String print2=setInfo(d4);
        //need to be updated
        textView1.setText(print1);
        textView2.setText(print2);
        int count=0;
        if(d3.equals("3"))
        {   count+=1;
            wa.setText("0");
        }else{
            wa.setText(d3);
        }
        if(d4.equals("3"))
        {    count+=1;
            ba.setText("0");
        }else{
```

```
            ba.setText(d4);
        }
        ou.setText(""+count);
        setImage(d3,d4);
        setButton(d3,d4,pos);
    }//floor3
    else  if(pos == 3){
        d1=setCheck(da1,dd1);
        d2=setCheck(da2,dd2);
        d3=setCheck(da3,dd3);
        d4=setCheck(da4,dd4);
        d5=setCheck(da5,dd5);
        d6=setCheck(da6,dd6);
        String print1=setInfo(d5);
        String print2=setInfo(d6);
        //need to be updated
        textView1.setText(print1);
        textView2.setText(print2);
        int count=0;
        if(d5.equals("3"))
        {   count+=1;
            wa.setText("0");
        }else{
            wa.setText(d5);
        }
        if(d6.equals("3"))
        {   count+=1;
            ba.setText("0");
        }else{
            ba.setText(d6);
        }
        ou.setText(""+count);
        setImage(d5,d6);
        setButton(d5,d6,pos);
    }
}
private String setCheck(String adm, String door) {
    //1=working
    //0=occupied
    if(adm.equals("") && door.equals("") ){
```

```
Toast.makeText(getApplicationContext(),adm+door,Toast.LENGTH_LONG).show();
        }else if(adm.equals("0") && door.equals("0"))
        {
            return "3";
        }else if(adm.equals("0") && door.equals("1")){
            return "3";
        }else if(adm.equals("1") && door.equals("0")){
            return "0";
        }else if(adm.equals("1") && door.equals("1")){
            return "1";
        }
        return "null-error";
    }
    private void setButton(String bt1, String bt2,int pos) {
        if(bt1.equals("0") || bt1.equals("3"))
        {
            b1.setEnabled(true);
            b1.setVisibility(View.VISIBLE);
        }else{
            b1.setEnabled(false);
            b1.setVisibility(View.INVISIBLE);
        }
        if(bt2.equals("0") || bt2.equals("3"))
        {
            b2.setEnabled((true));
            b2.setVisibility(View.VISIBLE);
        }else{
            b2.setEnabled(false);
            b2.setVisibility(View.INVISIBLE);
        }
        b1.setOnClickListener(new View.OnClickListener(){
            @Override
            public void onClick(View view) {
                //on floor1
                if(pos==1)
                {
                    if(d3.equals("1"))
                    {
```

```
                        Toast.makeText(getApplicationContext(),"Nearest
Washroom at Floor 2",Toast.LENGTH_LONG).show();
                        spinner.setSelection(2);
                    }else if(d5.equals("1"))
                    {
                        Toast.makeText(getApplicationContext(),"Nearest
Washroom at Floor 3",Toast.LENGTH_LONG).show();
                        spinner.setSelection(3);
                    }else{
                        Toast.makeText(getApplicationContext(),"No
Washrooms Available",Toast.LENGTH_LONG).show();
                    }
                }
                else if(pos==2)
                {
                    if(d1.equals("1"))
                    {
                        Toast.makeText(getApplicationContext(),"Nearest
Washroom at Floor 1",Toast.LENGTH_LONG).show();
                        spinner.setSelection(1);
                    }else if(d5.equals("1"))
                    {
                        Toast.makeText(getApplicationContext(),"Nearest
Washroom at Floor 3",Toast.LENGTH_LONG).show();
                        spinner.setSelection(3);
                    }else{
                        Toast.makeText(getApplicationContext(),"No
Washrooms Available",Toast.LENGTH_LONG).show();
                    }
                }
                else if(pos==3)
                {
                    if(d3.equals("1"))
                    {
                        Toast.makeText(getApplicationContext(),"Nearest
Washroom at Floor 2",Toast.LENGTH_LONG).show();
                        spinner.setSelection(2);
                    }else if(d1.equals("1"))
                    {
```

```
                        Toast.makeText(getApplicationContext(),"Nearest
Washroom at Floor 1",Toast.LENGTH_LONG).show();
                        spinner.setSelection(1);
                    }else{
                        Toast.makeText(getApplicationContext(),"No
Washrooms Available",Toast.LENGTH_LONG).show();
                    }
                }
            }
        });
        b2.setOnClickListener(new View.OnClickListener(){
            @Override
            public void onClick(View view) {
                //on floor1
                if(pos==1)
                {
                    if(d4.equals("1"))
                    {
                        Toast.makeText(getApplicationContext(),"Nearest
Bathroom at Floor 2",Toast.LENGTH_LONG).show();
                        spinner.setSelection(2);
                    }else if(d6.equals("1"))
                    {
                        Toast.makeText(getApplicationContext(),"Nearest
Bathroom at Floor 3",Toast.LENGTH_LONG).show();
                        spinner.setSelection(3);
                    }else{
                        Toast.makeText(getApplicationContext(),"No
Bathrooms Available",Toast.LENGTH_LONG).show();
                    }
                }
                else if(pos==2)
                {
                    if(d2.equals("1"))
                    {
                        Toast.makeText(getApplicationContext(),"Nearest
Bathroom at Floor 1",Toast.LENGTH_LONG).show();
                        spinner.setSelection(1);
                    }else if(d6.equals("1"))
                    {
```

```
                        Toast.makeText(getApplicationContext(),"Nearest
Bathroom at Floor 3",Toast.LENGTH_LONG).show();
                        spinner.setSelection(3);
                    }else{
                        Toast.makeText(getApplicationContext(),"No
Bathrooms Available",Toast.LENGTH_LONG).show();
                    }
                }
                else if(pos==3)
                {
                    if(d4.equals("1"))
                    {
                        Toast.makeText(getApplicationContext(),"Nearest
Bathroom at Floor 2",Toast.LENGTH_LONG).show();
                        spinner.setSelection(2);
                    }else if(d2.equals("1"))
                    {
                        Toast.makeText(getApplicationContext(),"Nearest
Bathroom at Floor 1",Toast.LENGTH_LONG).show();
                        spinner.setSelection(1);
                    }else{
                        Toast.makeText(getApplicationContext(),"No
Bathrooms Available",Toast.LENGTH_LONG).show();
                    }
                }
            }
        });
    }
    private String setInfo(String info){
        if(info.equals("1")){
            return "Vacant";
        }else  if(info.equals("0")){
            return "Occupied";
        }else  if(info.equals("3")){
            return "Out Of Service";
        }
        return "error";
    }
    private void setImage(String f1, String f2) {
        im1 = (ImageView)findViewById(R.id.imageView3);
```

```
        im2= (ImageView)findViewById(R.id.imageView2);
        if(f1.equals("0"))
        {
            im1.setImageResource(R.drawable.occuopiedw);
        }else if(f1.equals("3")){
            im1.setImageResource(R.drawable.ousw);
        }
        else {
            im1.setImageResource(R.drawable.vacantw);
        }
        if(f2.equals("0"))
        {
            im2.setImageResource(R.drawable.occupiedb);
        }else if(f2.equals("3")){
            im2.setImageResource(R.drawable.ousb);
        }
        else{
            im2.setImageResource(R.drawable.vacantb);
        }
    }
}
```

12.5 AndroidStudio (Admin App)

12.5.1 MainActivity.java

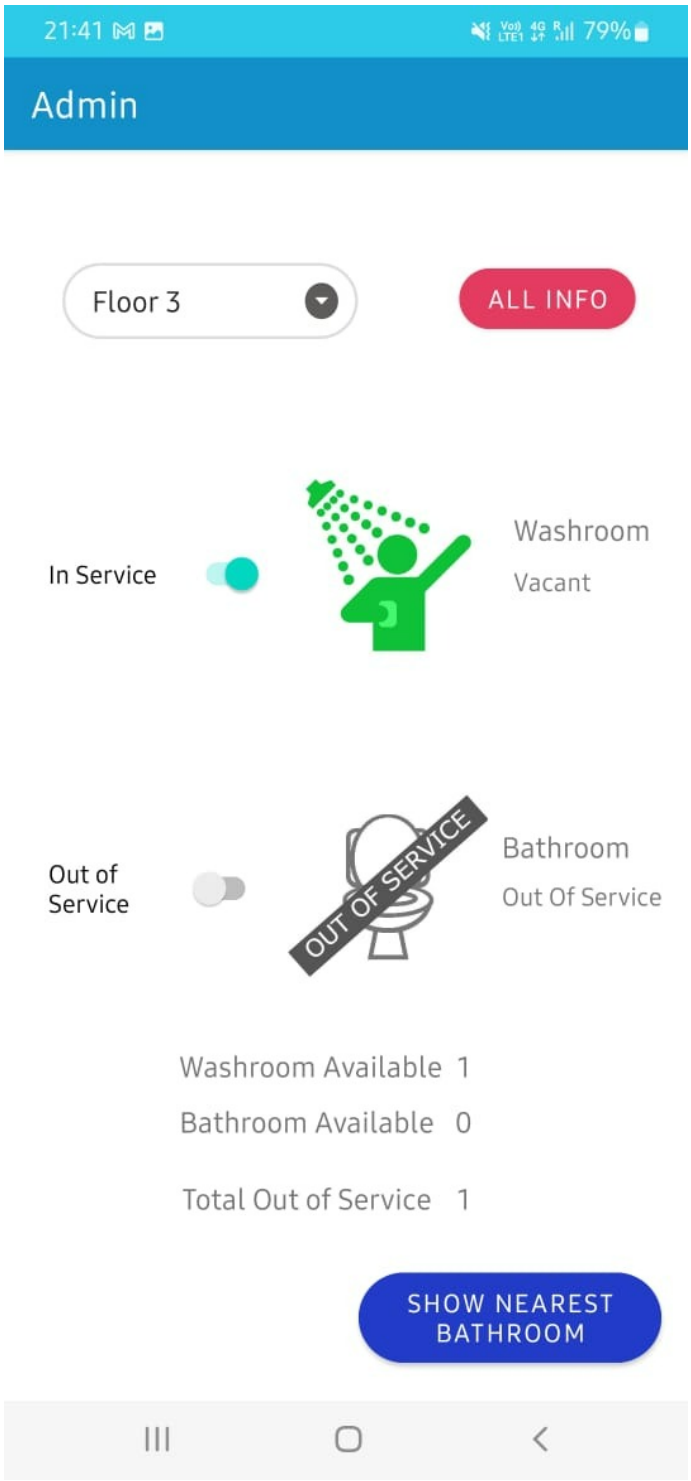


Fig 8-App Image

```java
package com.rixxx.myapplication;

import androidx.annotation.NonNull;
import androidx.annotation.RequiresApi;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.graphics.Color;
import android.os.Build;
import android.os.Bundle;
import android.os.Handler;
import android.util.Log;
import android.view.MotionEvent;
import android.view.View;
import android.view.ViewGroup;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.CompoundButton;
import android.widget.ImageView;
import android.widget.Spinner;
import android.widget.Switch;
import android.widget.TextView;
import android.widget.Toast;
import java.util.ArrayList;
import java.util.Arrays;
import java.util.HashMap;
import java.util.List;
import java.util.Map;
import com.android.volley.Request;
import com.android.volley.RequestQueue;
import com.android.volley.Response;
import com.android.volley.VolleyError;
import com.android.volley.toolbox.StringRequest;
import com.android.volley.toolbox.Volley;

public class MainActivity extends AppCompatActivity {

    TextView textView1;
    TextView textView2;
    Button b1,b2,inf;
```

```
    int count=0;
    TextView wa,ba,ou;
    RequestQueue queue,queue1,queue2,queue3;
    ImageView im1,im2;
    String d1,d2,d3,d4,d5,d6;
    Spinner spinner;
    String dd1,dd2,dd3,dd4,dd5,dd6;
    String da1,da2,da3,da4,da5,da6;
    String arr[]=new String[6];
    Switch select,s1,s2;
    String URL =
"https://api.thingspeak.com/channels/1897117/feeds.json?api_key=1L024NOIEU
XXUVSB&results=2";
    @RequiresApi(api = Build.VERSION_CODES.N)
    //adminside
    String
url2="https://api.thingspeak.com/channels/1911327/feeds.json?api_key=B1B7N
XKTIA3LQGK8&results=2";
    String
url="https://api.thingspeak.com/update?api_key=0MSY5CSW5B65MB5G";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        content();
        TextView inftxt1=(TextView)findViewById(R.id.info1);
        TextView inftxt2=(TextView)findViewById(R.id.info2);
        inf=findViewById(R.id.button2);
        inf.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                startActivity(new Intent(MainActivity.this,
MainActivity2.class));
            }
        });
        select =   findViewById(R.id.switch_select_all);
        s1 =  findViewById(R.id.switch1);
        s2 = (Switch)  findViewById(R.id.switch2);
        b1= findViewById(R.id.buttonw);
```

```
b2= (Button)findViewById(R.id.buttonb);
b1.setEnabled(false);
select.setEnabled(false);
s1.setEnabled(false);
s2.setEnabled(false);
select.setVisibility(View.INVISIBLE);
s1.setVisibility(View.INVISIBLE);
s2.setVisibility(View.INVISIBLE);
b1.setVisibility(View.INVISIBLE);
b2.setEnabled(false);
b2.setVisibility(View.INVISIBLE);
spinner = findViewById(R.id.spinner);
// Initializing a String Array
String[] plants = new String[]{
        "Select Floor...",
        "Floor 1",
        "Floor 2",
        "Floor 3",

};

List<String> plantsList = new ArrayList<>
        (Arrays.asList(plants));
ArrayAdapter<String> spinnerArrayAdapter
        = new ArrayAdapter<String>(
        this,
        android.R.layout.simple_dropdown_item_1line,
        plantsList
){
    @Override
    public boolean isEnabled(int position){
           setArray();
        return position != 0;
    }
    @Override
    public View getDropDownView(
            int position, View convertView,
            @NonNull ViewGroup parent) {
        View view = super.getDropDownView(
                position, convertView, parent);
```

```
                TextView textView = (TextView) view;
                if(position == 0){
                        textView.setTextColor(Color.GRAY);
                }
                  return view;
            }
        };

        spinnerArrayAdapter.setDropDownViewResource(
                android.R.layout.simple_dropdown_item_1line
        );

        spinner.setOnItemSelectedListener(
                new AdapterView.OnItemSelectedListener() {
                  @Override
                    public void onItemSelected(
                            AdapterView<?> parent, View view,
                            int position, long id) {

                        // Get the spinner selected item text
                        String selectedItemText = (String) parent
                                .getItemAtPosition(position);

                        textView1 = findViewById(R.id.info1);
                        textView2 = findViewById(R.id.info2);
                        wa = findViewById(R.id.wav);
                        ba = findViewById(R.id.bav);
                        ou = findViewById(R.id.ous);
                        if(position == 1){
                             textSwitch();

Toast.makeText(getApplicationContext(),"Selected : "+
selectedItemText,Toast.LENGTH_SHORT).show();
                            d1=setCheck(arr[0],dd1);
                            d2=setCheck(arr[1],dd2);
                            d3=setCheck(arr[2],dd3);
                            d4=setCheck(arr[3],dd4);
                            d5=setCheck(arr[4],dd5);
                            d6=setCheck(arr[5],dd6);
                            String print1=setInfo(d1);
```

```
                    String print2=setInfo(d2);
                    textView1.setText(print1);
                    textView2.setText(print2);
                    int count=0;
                    if(d1.equals("3"))
                    {     count+=1;
                        wa.setText("0");
                    }else{
                        wa.setText(d1);
                    }
                    if(d2.equals("3"))
                    {   count+=1;
                        ba.setText("0");
                    }else{
                        ba.setText(d2);
                    }
                    ou.setText(""+count);
                    setSwitch(d1,d2,position);
                    setButton(d1,d2,position);
                    setImage(d1,d2);
                }//floor2
                else  if(position == 2){
                     textSwitch();

Toast.makeText(getApplicationContext(),"Selected : "+
selectedItemText,Toast.LENGTH_SHORT).show();
                    d1=setCheck(arr[0],dd1);
                    d2=setCheck(arr[1],dd2);
                    d3=setCheck(arr[2],dd3);
                    d4=setCheck(arr[3],dd4);
                    d5=setCheck(arr[4],dd5);
                    d6=setCheck(arr[5],dd6);
                    String print1=setInfo(d3);
                    String print2=setInfo(d4);
                    textView1.setText(print1);
                    textView2.setText(print2);
                    int count=0;
                    if(d3.equals("3"))
                    {    count+=1;
                        wa.setText("0");
```

```
                    }else{
                        wa.setText(d3);
                    }
                    if(d4.equals("3"))
                    {
                        count+=1;
                        ba.setText("0");
                    }else{
                        ba.setText(d4);
                    }
                    ou.setText(""+count);
                    setSwitch(d3,d4,position);
                    setImage(d3,d4);
                    setButton(d3,d4,position);
                }//floor3
                else  if(position == 3){
                      textSwitch();

Toast.makeText(getApplicationContext(),"Selected : "+
selectedItemText,Toast.LENGTH_SHORT).show();
                    d1=setCheck(arr[0],dd1);
                    d2=setCheck(arr[1],dd2);
                    d3=setCheck(arr[2],dd3);
                    d4=setCheck(arr[3],dd4);
                    d5=setCheck(arr[4],dd5);
                    d6=setCheck(arr[5],dd6);
                    String print1=setInfo(d5);
                    String print2=setInfo(d6);
                   textView1.setText(print1);
                    textView2.setText(print2);
                    int count=0;
                    if(d5.equals("3"))
                    {   count+=1;
                        wa.setText("0");
                    }else{
                        wa.setText(d5);
                    }
                    if(d6.equals("3"))
                    {
                       count+=1;
```

```
                                ba.setText("0");
                            }else{
                                ba.setText(d6);
                            }
                            ou.setText(""+count);
                            setSwitch(d5,d6,position);
                            setImage(d5,d6);
                            setButton(d5,d6,position);
                        }
                    }
                    @Override
                    public void onNothingSelected(
                            AdapterView<?> parent) {
                    }
                });
         spinner.setAdapter(spinnerArrayAdapter);
        contData();
        s1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                String text1 =
Integer.toString(spinner.getSelectedItemPosition());
               String door_admin1,door_admin2;
                if(s1.isChecked()==true)
                {
                    s1.setText("In Service");
                    door_admin1="1";
                }else{
                    s1.setText("Out of Service");
                    door_admin1="0";
                }
                setAdminData1(text1,door_admin1);
                write_data(arr[0],arr[1],arr[2],arr[3],arr[4],arr[5]);

Toast.makeText(getApplicationContext(),"Updating",Toast.LENGTH_LONG).show(
);
            }
         });
         s2.setOnClickListener(new View.OnClickListener() {
            @Override
```

```
            public void onClick(View view) {
                String text2 =
Integer.toString(spinner.getSelectedItemPosition());
            String door_admin1,door_admin2;
                if(s2.isChecked()==true)
                {   s2.setText("In Service");
                    door_admin2="1";
                }else{
                    s2.setText("Out of Service");
                    door_admin2="0";
                }
               setAdminData2(text2,door_admin2);
                write_data(arr[0],arr[1],arr[2],arr[3],arr[4],arr[5]);

Toast.makeText(getApplicationContext(),"Updating",Toast.LENGTH_LONG).show(
);
            }
        });
   }
   private void setArray() {
        arr[0]=da1;
        arr[1]=da2;
        arr[2]=da3;
        arr[3]=da4;
        arr[4]=da5;
        arr[5]=da6;
    }
    public void content() {
        queue = Volley.newRequestQueue(this);
        StringRequest request = new StringRequest(Request.Method.GET, URL,
new Response.Listener<String>() {
            @Override
            public void onResponse(String response) {
                String s=response.toString();
                dd1=Character.toString(s.charAt(s.length()-70));
                dd2=Character.toString(s.charAt(s.length()-57));
                dd3=Character.toString(s.charAt(s.length()-44));
                dd4=Character.toString(s.charAt(s.length()-31));
                dd5=Character.toString(s.charAt(s.length()-18));
                dd6=Character.toString(s.charAt(s.length()-5));
```

```
            }
        }, new Response.ErrorListener() {
            @Override
            public void onErrorResponse(VolleyError error) {
                Log.d("error",error.toString());
            }
        });
        queue.add(request);
        //adminside
        queue1 = Volley.newRequestQueue(this);
        StringRequest request1 = new StringRequest(Request.Method.GET,
url2, new Response.Listener<String>() {
            @Override
            public void onResponse(String response) {
                String sk=response.toString();
                da1=Character.toString(sk.charAt(sk.length()-70));
                da2=Character.toString(sk.charAt(sk.length()-57));
                da3=Character.toString(sk.charAt(sk.length()-44));
                da4=Character.toString(sk.charAt(sk.length()-31));
                da5=Character.toString(sk.charAt(sk.length()-18));
                da6=Character.toString(sk.charAt(sk.length()-5));
            }
        }, new Response.ErrorListener() {
            @Override
            public void onErrorResponse(VolleyError error) {
                Log.d("error",error.toString());
            }
        });
        queue1.add(request1);
       refresh1(4000);
    }
    private void refresh1(int i) {
        final Handler handler=new Handler();
        final Runnable runnable=new Runnable() {
            @Override
            public void run() {
                content();
                setArray();
                contData();
             }
```

```
        };
        handler.postDelayed(runnable,i);
    }
    private void setAdminData1(String f1, String dad1) {
        if (f1.equals("1"))
        {
            arr[0]=dad1;
        }else if(f1.equals("2")){
            arr[2]=dad1;
        }else if(f1.equals("3")){
            arr[4]=dad1;
        }
    }
    private void setAdminData2(String f1, String dad2) {
        if (f1.equals("1"))
        {
           arr[1]=dad2;
        }else if(f1.equals("2")){
           arr[3]=dad2;
        }else if(f1.equals("3")){
           arr[5]=dad2;
        }
    }
    private void write_data(String wd1, String wd2, String wd3,String
wd4,String wd5,String wd6) {
        queue3 = Volley.newRequestQueue(this);
        StringRequest putRequest = new StringRequest(Request.Method.POST,
url,
                new Response.Listener<String>()
                {
                    @Override
                    public void onResponse(String response) {
                        // response
                        Log.d("Response", response);
                    }
                },
                new Response.ErrorListener()
                {
                    @Override
                    public void onErrorResponse(VolleyError error) {
```

```
                        // error
                        Log.d("Error.Response", error.toString());
                    }
                }
        ) {

            @Override
            protected Map<String, String> getParams()
            {
                Map<String, String>  params = new HashMap<String,
String>();
                params.put("field1",wd1);
                params.put("field2",wd2);
                params.put("field3", wd3);
                params.put("field4", wd4);
                params.put("field5", wd5);
                params.put("field6", wd6);
                return params;
            }
        };
        queue3.add(putRequest);
        arr[0]=wd1;
        arr[1]=wd2;
        arr[2]=wd3;
        arr[3]=wd4;
        arr[4]=wd4;
        arr[5]=wd5;
      contData();
    }
     private void contData(){
        int pos=spinner.getSelectedItemPosition();
        textView1 = findViewById(R.id.info1);
        textView2 = findViewById(R.id.info2);
        wa = findViewById(R.id.wav);
        ba = findViewById(R.id.bav);
        ou = findViewById(R.id.ous);
        if(pos == 1){
              textSwitch();

            d1=setCheck(arr[0],dd1);
```

```
        d2=setCheck(arr[1],dd2);
        d3=setCheck(arr[2],dd3);
        d4=setCheck(arr[3],dd4);
        d5=setCheck(arr[4],dd5);
        d6=setCheck(arr[5],dd6);
        String print1=setInfo(d1);
        String print2=setInfo(d2);
        textView1.setText(print1);
        textView2.setText(print2);
        int count=0;
        if(d1.equals("3"))
        {   count+=1;
            wa.setText("0");
        }else{
            wa.setText(d1);
        }
        if(d2.equals("3"))
        {
            count+=1;
            ba.setText("0");
        }else{
            ba.setText(d2);
        }
        ou.setText(""+count);
        setButton(d1,d2,pos);
        setImage(d1,d2);
    }//floor2
    else  if(pos == 2){
         textSwitch();
        d1=setCheck(arr[0],dd1);
        d2=setCheck(arr[1],dd2);
        d3=setCheck(arr[2],dd3);
        d4=setCheck(arr[3],dd4);
        d5=setCheck(arr[4],dd5);
        d6=setCheck(arr[5],dd6);
        String print1=setInfo(d3);
        String print2=setInfo(d4);
        textView1.setText(print1);
        textView2.setText(print2);
        int count=0;
```

```
        if(d3.equals("3"))
        {
            count+=1;
            wa.setText("0");
        }else{
            wa.setText(d3);
        }
        if(d4.equals("3"))
        {   count+=1;
            ba.setText("0");
        }else{
            ba.setText(d4);
        }
        ou.setText(""+count);
        setImage(d3,d4);
        setButton(d3,d4,pos);
    }//floor3
    else  if(pos == 3){
          textSwitch();
        d1=setCheck(arr[0],dd1);
        d2=setCheck(arr[1],dd2);
        d3=setCheck(arr[2],dd3);
        d4=setCheck(arr[3],dd4);
        d5=setCheck(arr[4],dd5);
        d6=setCheck(arr[5],dd6);
        String print1=setInfo(d5);
        String print2=setInfo(d6);
        //need to be updated
        textView1.setText(print1);
        textView2.setText(print2);
        int count=0;
        if(d5.equals("3"))
        {   count+=1;
            wa.setText("0");
        }else{
            wa.setText(d5);
        }
        if(d6.equals("3"))
        {   count+=1;
            ba.setText("0");
```

```
            }else{
                ba.setText(d6);
            }
            ou.setText(""+count);
            setImage(d5,d6);
            setButton(d5,d6,pos);
        }
    }
    private String setCheck(String adm, String door) {
        //1=working
        //0=occupied
        if(adm.equals("") && door.equals("") ){
           //
Toast.makeText(getApplicationContext(),adm+door,Toast.LENGTH_LONG).show();
        }else if(adm.equals("0") && door.equals("0"))
        {
            return "3";
        }else if(adm.equals("0") && door.equals("1")){
            return "3";
        }else if(adm.equals("1") && door.equals("0")){
            return "0";
        }else if(adm.equals("1") && door.equals("1")){
            return "1";
        }
        return "null-error";
    }
    private void setButton(String bt1, String bt2,int pos) {
        if(bt1.equals("0") || bt1.equals("3"))
        {
            b1.setEnabled(true);
            b1.setVisibility(View.VISIBLE);
        }else{
            b1.setEnabled(false);
            b1.setVisibility(View.INVISIBLE);
        }
        if(bt2.equals("0") || bt2.equals("3"))
        {
            b2.setEnabled((true));
            b2.setVisibility(View.VISIBLE);
        }else{
```

```
            b2.setEnabled(false);
            b2.setVisibility(View.INVISIBLE);
        }
        b1.setOnClickListener(new View.OnClickListener(){
            @Override
            public void onClick(View view) {
                //on floor1
                if(pos==1)
                {
                    if(d3.equals("1"))
                    {
                        Toast.makeText(getApplicationContext(),"Nearest
Washroom at Floor 2",Toast.LENGTH_SHORT).show();
                        spinner.setSelection(2);
                    }else if(d5.equals("1"))
                    {
                        Toast.makeText(getApplicationContext(),"Nearest
Washroom at Floor 3",Toast.LENGTH_SHORT).show();
                        spinner.setSelection(3);
                    }else{
                        Toast.makeText(getApplicationContext(),"No
Washrooms Available",Toast.LENGTH_SHORT).show();
                    }
                }
                else if(pos==2)
                {
                    if(d1.equals("1"))
                    {
                        Toast.makeText(getApplicationContext(),"Nearest
Washroom at Floor 1",Toast.LENGTH_SHORT).show();
                        spinner.setSelection(1);
                    }else if(d5.equals("1"))
                    {
                        Toast.makeText(getApplicationContext(),"Nearest
Washroom at Floor 3",Toast.LENGTH_SHORT).show();
                        spinner.setSelection(3);
                    }else{
                        Toast.makeText(getApplicationContext(),"No
Washrooms Available",Toast.LENGTH_SHORT).show();
                    }
```

```
                }
                else if(pos==3)
                {
                    if(d3.equals("1"))
                    {
                        Toast.makeText(getApplicationContext(),"Nearest
Washroom at Floor 2",Toast.LENGTH_SHORT).show();
                        spinner.setSelection(2);


                    }else if(d1.equals("1"))
                    {
                        Toast.makeText(getApplicationContext(),"Nearest
Washroom at Floor 1",Toast.LENGTH_SHORT).show();
                        spinner.setSelection(1);

                    }else{
                        Toast.makeText(getApplicationContext(),"No
Washrooms Available",Toast.LENGTH_SHORT).show();
                    }
                }
            }
        });
        b2.setOnClickListener(new View.OnClickListener(){
            @Override
            public void onClick(View view) {
                //on floor1
                if(pos==1)
                {
                    if(d4.equals("1"))
                    {
                        Toast.makeText(getApplicationContext(),"Nearest
Bathroom at Floor 2",Toast.LENGTH_SHORT).show();
                        spinner.setSelection(2);
                    }else if(d6.equals("1"))
                    {
                        Toast.makeText(getApplicationContext(),"Nearest
Bathroom at Floor 3",Toast.LENGTH_SHORT).show();
                        spinner.setSelection(3);
                    }else{
```

```
                        Toast.makeText(getApplicationContext(),"No
Bathrooms Available",Toast.LENGTH_SHORT).show();
                    }
                }
                else if(pos==2)
                {
                    if(d2.equals("1"))
                    {
                        Toast.makeText(getApplicationContext(),"Nearest
Bathroom at Floor 1",Toast.LENGTH_SHORT).show();
                        spinner.setSelection(1);
                    }else if(d6.equals("1"))
                    {
                        Toast.makeText(getApplicationContext(),"Nearest
Bathroom at Floor 3",Toast.LENGTH_SHORT).show();
                        spinner.setSelection(3);
                    }else{
                        Toast.makeText(getApplicationContext(),"No
Bathrooms Available",Toast.LENGTH_SHORT).show();
                    }
                }
                else if(pos==3)
                {
                    if(d4.equals("1"))
                    {
                        Toast.makeText(getApplicationContext(),"Nearest
Bathroom at Floor 2",Toast.LENGTH_SHORT).show();
                        spinner.setSelection(2);
                    }else if(d2.equals("1"))
                    {
                        Toast.makeText(getApplicationContext(),"Nearest
Bathroom at Floor 1",Toast.LENGTH_SHORT).show();
                        spinner.setSelection(1);
                    }else{
                        Toast.makeText(getApplicationContext(),"No
Bathrooms Available",Toast.LENGTH_SHORT).show();
                    }
                }
            }
        });
```

```
}
private void setImage(String f1, String f2) {
    im1 = (ImageView)findViewById(R.id.imageView3);
    im2= (ImageView)findViewById(R.id.imageView2);
    if(f1.equals("0"))
    {
        im1.setImageResource(R.drawable.occuopiedw);
    }else if(f1.equals("3")){
        im1.setImageResource(R.drawable.ousw);
    }
    else {
        im1.setImageResource(R.drawable.vacantw);
    }
    if(f2.equals("0"))
    {
        im2.setImageResource(R.drawable.occupiedb);
    }else if(f2.equals("3")){
        im2.setImageResource(R.drawable.ousb);
    }
    else{
        im2.setImageResource(R.drawable.vacantb);
    }
}
private  void setSwitch(String a1,String a2,int pos){
    if(!a1.equals("3"))
    {
        s1.setChecked(true);
    }else{
        s1.setChecked(false);
    }
    if(!a2.equals("3"))
    {
        s2.setChecked(true);
    }else{
        s2.setChecked(false);
    }

    if(!a1.equals("3") && !a2.equals("3"))
    {
      //  select.setChecked(true);
```

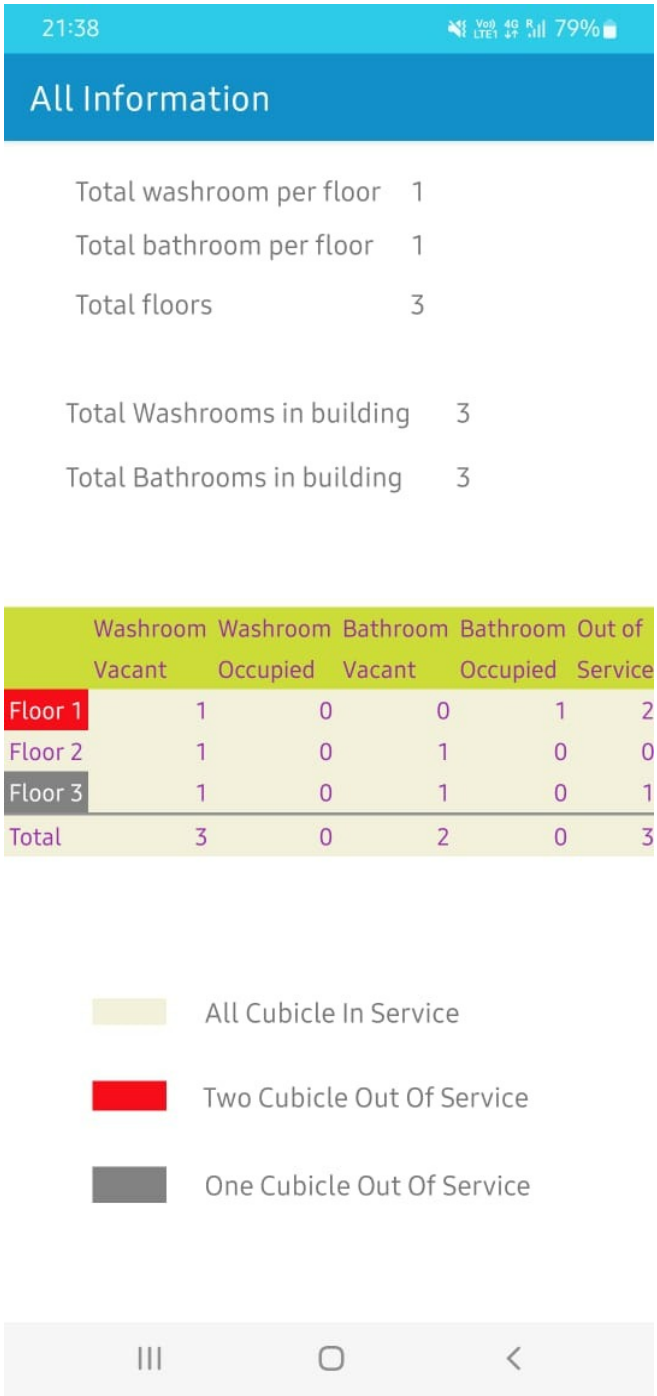
```
    }else{
        if(s1.isChecked()==true )
        {
           // select.setChecked(false);
            s1.setChecked(true);
        }
        if(s2.isChecked()==true )
        {
           // select.setChecked(false);
            s2.setChecked(true);
        }
    }
    if(pos>0)
    {
       // select.setEnabled(true);
        s1.setEnabled(true);
        s2.setEnabled(true);
      //  select.setVisibility(View.VISIBLE);
        s1.setVisibility(View.VISIBLE);
        s2.setVisibility(View.VISIBLE);
    }else{
      //  select.setEnabled(false);
        s1.setEnabled(false);
        s2.setEnabled(false);
      //  select.setVisibility(View.INVISIBLE);
        s1.setVisibility(View.INVISIBLE);
        s2.setVisibility(View.INVISIBLE);
    }
}
private void textSwitch() {

    if(s1.isChecked()==true)
    {
        s1.setText("In Service");
    }else
    {
        s1.setText("Out of Service");
    }
    if(s2.isChecked()==true)
    {
```

```
            s2.setText("In Service");
        }else
        {
            s2.setText("Out of Service");
        }

    }
    private String setInfo(String info){
        if(info.equals("1")){
            return "Vacant";
        }else  if(info.equals("0")){
            return "Occupied";
        }else  if(info.equals("3")){
            return "Out Of Service";
        }
        return "error";
    }
}
```

12.5.2 MainActivity2.java(All-info)



	Washroom Vacant	Washroom Occupied	Bathroom Vacant	Bathroom Occupied	Out of Service
Floor 1	1	0	0	1	2
Floor 2	1	0	1	0	0
Floor 3	1	0	1	0	1
Total	3	0	2	0	3

Fig 9-App Image

```java
package com.rixxx.myapplication;

import androidx.annotation.RequiresApi;
import androidx.appcompat.app.AppCompatActivity;
import android.graphics.Color;
import android.os.Build;
import android.os.Bundle;
import android.os.Handler;
import android.util.Log;
import android.view.View;
import android.widget.NumberPicker;
import android.widget.TextView;
import android.widget.Toast;
import com.android.volley.Request;
import com.android.volley.RequestQueue;
import com.android.volley.Response;
import com.android.volley.VolleyError;
import com.android.volley.toolbox.StringRequest;
import com.android.volley.toolbox.Volley;

public class MainActivity2 extends AppCompatActivity {
    RequestQueue queue,queue1,queue2,queue3;
    String dd1,dd2,dd3,dd4,dd5,dd6;
    String arr[]=new String[6];
    String da1,da2,da3,da4,da5,da6;
    String URL =
"https://api.thingspeak.com/channels/1897117/feeds.json?api_key=1L024NOIEU
XXUVSB&results=2";
    @RequiresApi(api = Build.VERSION_CODES.N)
    //adminside
    String
url2="https://api.thingspeak.com/channels/1911327/feeds.json?api_key=B1B7N
XKTIA3LQGK8&results=2";
    String
url="https://api.thingspeak.com/update?api_key=0MSY5CSW5B65MB5G";
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2);
        setTitle("All Information");
```

```
        content();
    }
    private void content() {
       queue = Volley.newRequestQueue(this);
        StringRequest request = new StringRequest(Request.Method.GET, URL,
new Response.Listener<String>() {
            @Override
            public void onResponse(String response) {
                String s=response.toString();
                dd1=Character.toString(s.charAt(s.length()-70));
                dd2=Character.toString(s.charAt(s.length()-57));
                dd3=Character.toString(s.charAt(s.length()-44));
                dd4=Character.toString(s.charAt(s.length()-31));
                dd5=Character.toString(s.charAt(s.length()-18));
                dd6=Character.toString(s.charAt(s.length()-5));
                TextView w1v=(TextView) findViewById(R.id.w1v);
                TextView w1o=(TextView) findViewById(R.id.w1o);
                TextView b1v=(TextView) findViewById(R.id.b1v);
                TextView b1o=(TextView) findViewById(R.id.b1o);
                TextView w2v=(TextView) findViewById(R.id.w2v);
                TextView w2o=(TextView) findViewById(R.id.w2o);
                TextView b2v=(TextView) findViewById(R.id.b2v);
                TextView b2o=(TextView) findViewById(R.id.b2o);
                TextView w3v=(TextView) findViewById(R.id.w3v);
                TextView w3o=(TextView) findViewById(R.id.w3o);
                TextView b3v=(TextView) findViewById(R.id.b3v);
                TextView b3o=(TextView) findViewById(R.id.b3o);
                TextView twv=(TextView) findViewById(R.id.twv);
                TextView two=(TextView) findViewById(R.id.two);
                TextView tbv=(TextView) findViewById(R.id.tbv);
                TextView tbo=(TextView) findViewById(R.id.tbo);
                Integer n1=Integer.parseInt(dd1);
                Integer n2=Integer.parseInt(dd2);
                Integer n3=Integer.parseInt(dd3);
                Integer n4=Integer.parseInt(dd4);
                Integer n5=Integer.parseInt(dd5);
                Integer n6=Integer.parseInt(dd6);
                if(n1==1)
                {
                    w1v.setText("1");
```

```
        w1o.setText("0");
    }else{
        w1v.setText("0");
        w1o.setText("1");
    }
    if(n2==1)
    {
        b1v.setText("1");
        b1o.setText("0");
    }else{
        b1v.setText("0");
        b1o.setText("1");
    }
    if(n3==1)
    {
        w2v.setText("1");
        w2o.setText("0");
    }else{
        w2v.setText("0");
        w2o.setText("1");
    }
    if(n4==1)
    {
        b2v.setText("1");
        b2o.setText("0");
    }else{
        b2v.setText("0");
        b2o.setText("1");
    }
    if(n5==1)
    {
        w3v.setText("1");
        w3o.setText("0");
    }else{
        w3v.setText("0");
        w3o.setText("1");
    }
    if(n6==1)
    {
        b3v.setText("1");
```

```
                    b3o.setText("0");
                }else{
                    b3v.setText("0");
                    b3o.setText("1");
                }
                int sum1=n1+n3+n5;
                int sum2=3-sum1;
                twv.setText(""+sum1);
                two.setText(""+sum2);
                int sum3=n2+n4+n6;
                int sum4=3-sum1;
                tbv.setText(""+sum3);
                tbo.setText(""+sum4);
            }
        }, new Response.ErrorListener() {
            @Override
            public void onErrorResponse(VolleyError error) {
                Log.d("error",error.toString());
            }
        });
        queue.add(request);
        //adminside
        queue1 = Volley.newRequestQueue(this);
        StringRequest request1 = new StringRequest(Request.Method.GET,
url2, new Response.Listener<String>() {
            @Override
            public void onResponse(String response) {
                String sk=response.toString();
                da1=Character.toString(sk.charAt(sk.length()-70));
                da2=Character.toString(sk.charAt(sk.length()-57));
                da3=Character.toString(sk.charAt(sk.length()-44));
                da4=Character.toString(sk.charAt(sk.length()-31));
                da5=Character.toString(sk.charAt(sk.length()-18));
                da6=Character.toString(sk.charAt(sk.length()-5));
                TextView os1=(TextView) findViewById(R.id.os1);
                TextView os2=(TextView) findViewById(R.id.os2);
                TextView os3=(TextView) findViewById(R.id.os3);
                TextView tos=(TextView) findViewById(R.id.tos);
                TextView floor1=(TextView) findViewById(R.id.floor1);
                TextView floor2=(TextView) findViewById(R.id.floor2);
```

```
                TextView floor3=(TextView) findViewById(R.id.floor3);
                int k1=Integer.parseInt(da1);
                int k2=Integer.parseInt(da2);
                int k3=Integer.parseInt(da3);
                int k4=Integer.parseInt(da4);
                int k5=Integer.parseInt(da5);
                int k6=Integer.parseInt(da6);
                k1=k1==1?0:1;
                k2=k2==1?0:1;
                k3=k3==1?0:1;
                k4=k4==1?0:1;
                k5=k5==1?0:1;
                k6=k6==1?0:1;
                int d1=k1+k2;
                int d2=k3+k4;
                int d3=k5+k6;
                if(d1==0)
                {

floor1.setBackgroundColor(Color.parseColor("#F3F0DC"));
                    floor1.setTextColor(Color.parseColor("#9C27B0"));
                }
                else if(d1==1)
                {

floor1.setBackgroundColor(Color.parseColor("#828282"));
                    floor1.setTextColor(Color.parseColor("#ffffff"));
                }else if(d1==2)
                {

floor1.setBackgroundColor(Color.parseColor("#f70d1a"));
                    floor1.setTextColor(Color.parseColor("#ffffff"));
                }
                if(d2==0)
                {

floor2.setBackgroundColor(Color.parseColor("#F3F0DC"));
                    floor2.setTextColor(Color.parseColor("#9C27B0"));
                }
                else if(d2==1)
```

```
                {

floor2.setBackgroundColor(Color.parseColor("#828282"));
                    floor2.setTextColor(Color.parseColor("#ffffff"));
                }else if(d2==2)
                {

floor2.setBackgroundColor(Color.parseColor("#f70d1a"));
                    floor2.setTextColor(Color.parseColor("#ffffff"));
                }
                if(d3==0)
                {

floor3.setBackgroundColor(Color.parseColor("#F3F0DC"));
                    floor3.setTextColor(Color.parseColor("#9C27B0"));
                }
                else if(d3==1)
                {

floor3.setBackgroundColor(Color.parseColor("#828282"));
                    floor3.setTextColor(Color.parseColor("#ffffff"));
                }else if(d3==2)
                {

floor3.setBackgroundColor(Color.parseColor("#f70d1a"));
                    floor3.setTextColor(Color.parseColor("#ffffff"));
                }
                os1.setText(""+d1);
                os2.setText(""+d2);
                os3.setText(""+d3);
                int summ=d1+d2+d3;
                tos.setText(""+summ);
            }
        }, new Response.ErrorListener() {
            @Override
            public void onErrorResponse(VolleyError error) {
                Log.d("error",error.toString());
            }
        });
        queue1.add(request1);
```

```
        refresh1(1000);
    }
    private void refresh1(int i) {
        final Handler handler=new Handler();
        final Runnable runnable=new Runnable() {
            @Override
            public void run() {
                content();
                setArray();
            }
        };
        handler.postDelayed(runnable,i);
    }
    private void setArray() {
        arr[0]=da1;
        arr[1]=da2;
        arr[2]=da3;
        arr[3]=da4;
        arr[4]=da5;
        arr[5]=da6;
    }
    private String setCheck(String adm, String door) {
        //1=working
        //0=occupied
        if(adm.equals("") && door.equals("") ){

Toast.makeText(getApplicationContext(),adm+door,Toast.LENGTH_LONG).show();
        }else if(adm.equals("0") && door.equals("0"))
        {
            return "3";
        }else if(adm.equals("0") && door.equals("1")){
            return "3";
        }else if(adm.equals("1") && door.equals("0")){
            return "0";
        }else if(adm.equals("1") && door.equals("1")){
            return "1";
        }
        return "null-error";
    }
}
```

14. Conclusion

Since students always need to save their time whenever they can, waiting for washroom cubicles could be a real time killer. Our solution helps the person see the available cubicle in real time so not much time is wasted. This increases their productivity in other tasks.
Our app is easy to use and does not involve any complexities that diminishes the user experience.
This solution also helps solve the user frustration, as they would not have to check every cubicle physically to see if it is available.
Our solution is well optimized for usability and findability.Our app tries to remove ergonomic designs to optimize the user experience.

15. References

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